



5. Introduction to SPSS

Dr. Dr. Albulene Grajcevc

Today's agenda

- Measurement scales
- Introduction to SPSS
- Coding variables in SPSS
- EXCEL to SPSS
- Missing values
- Activities
 - Recode variables
 - Compute new variable

Measurement levels

- Nominal
 - Only used to classify data, no value assigned
- Ordinal
 - Implies an ordered relationship
- Ratio (continuous)
 - Equal intervals, but including a 0
- Interval/Scale (continuous)
 - Equal intervals

Measurement levels

- Nominal
 - Gender, location, hair color
- Ordinal
 - Income, education level
- Ratio
 - Temperature, points in an exam
- Interval/Scale
 - agreement, preference

Activity 1

- Type of movie you prefer
- The quality of service you receive at a restaurant
- Amount of protein in your drink
- Calories in a cookie
- Your score in an exam
- Time it takes to finish an exam

SPSS: INTRO

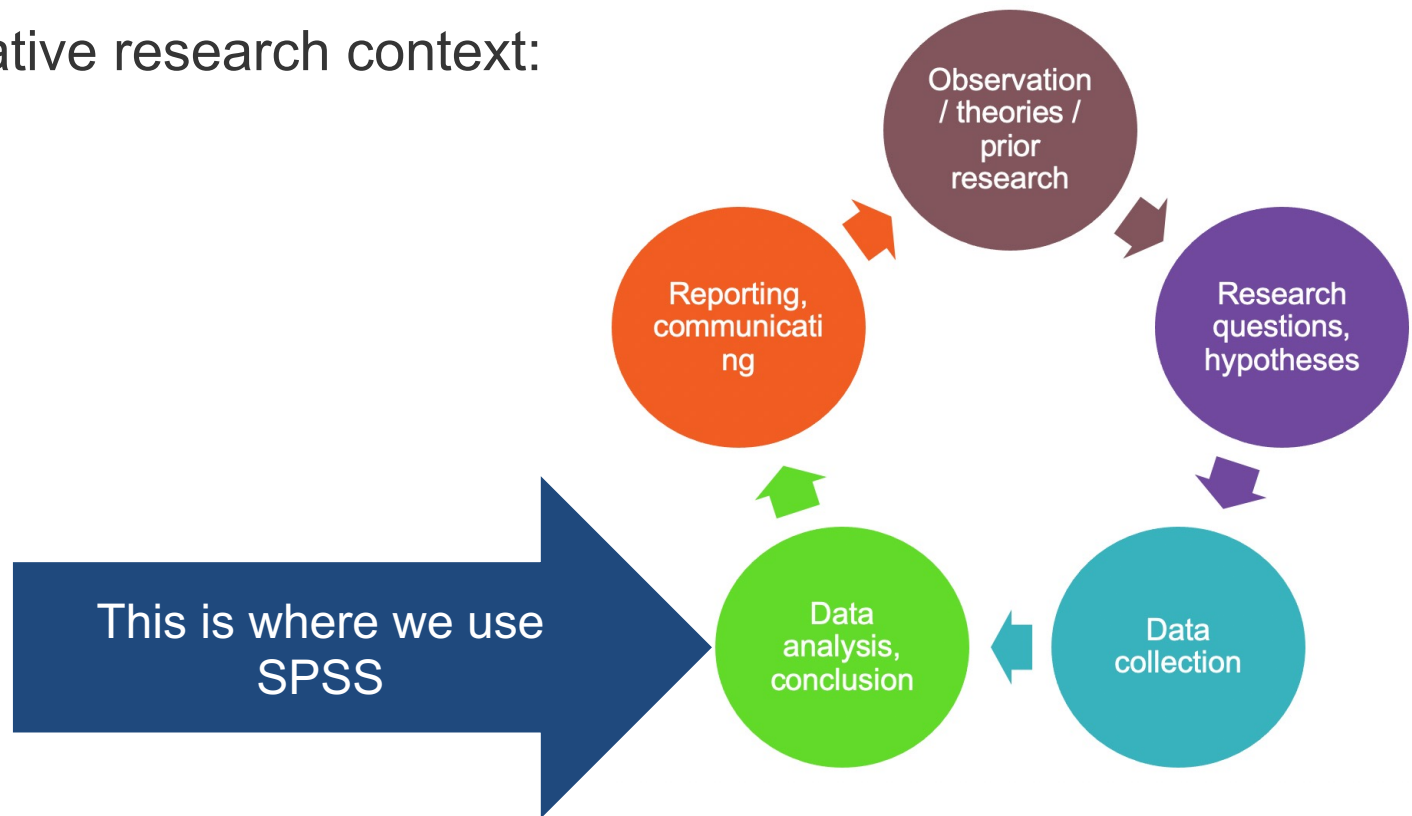
What is SPSS Statistics?

- ◇ Statistical Package for the Social Sciences
- ◇ Officially named "IBM SPSS Statistics"
- ◇ Current version: 27 (Windows / macOS / Linux)



What is SPSS Statistics used for?

- ◇ Manipulation and statistical analysis of survey data
- ◇ In a quantitative research context:



SPSS: SPSS ENVIRONMENT

SPSS Environment

Each SPSS window is associated with specific tasks and types of SPSS files.

Visible: 12 of 12 Variables

	F1_Media	F2_Ind_Emo	Exp_Condition	Gender	Age	Res_pre	Res_post	Mot_01	Mot_02	Mot_03
1	1	1	1	1	33	67.00	69.00	3	4	
2	1	1	1	2	20	79.00	81.00	4	1	
3	1	1	1	2	19	65.00	66.00	3	3	
4	1	1	1	2	23	89.00	79.00	5	4	
5	1	2	2	1	25	21.00	30.00	1	1	
6	1	2	2	2	20	62.00	76.00	3	3	
7	1	2	2	1	22	34.00	25.00	2	1	
8	1	2	2	1	27	54.00	30.00	1	1	
9	2	1	3	2	23	42.00	79.00	4	5	
10	2	1	3	1	24	58.00	77.00	5	5	
11	2	1	3	1	30	37.00	56.00	3	2	
12	2	1	3	2	26	97.00	100.00	5	4	
13	2	2	4	1	25	28.00	54.00	3	4	
14	2	2	4	1	28	41.00	45.00	3	3	
15	2	2	4	1	21	69.00	53.00	4	3	
16	2	2	4	1	24	58.00	33.00	2	1	

Data View Variable View

IBM SPSS Statistics Processor is ready Unicode:ON

Data Editor
window:
Data View

Input and see all your collected
data

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	M
1	F1_Media	Numeric	8	0	Factor 1: Learn...	{1, Tablet}...	None	8	Right	Nc
2	F2_Ind_Emo	Numeric	8	0	Factor 2: Induc...	{1, Joy}...	None	8	Right	Nc
3	Exp_Conditi...	Numeric	8	0	Experimental C...	{1, Tablet+}...	None	8	Right	Nc
4	Gender	Numeric	8	0	Gender	{1, Male}...	None	8	Right	Nc
5	Age	Numeric	8	0	Age		None	8	Right	Sc
6	Res_pre	Numeric	8	2	Result Pretest	None	None	8	Right	Sc
7	Res_post	Numeric	8	2	Result Posttest	None	None	8	Right	Sc
8	Mot_01	Numeric	8	0	Motivation Item...	None	None	8	Right	Sc
9	Mot_02	Numeric	8	0	Motivation Item...	None	None	8	Right	Sc
10	Mot_03	Numeric	8	0	Motivation Item...	None	None	8	Right	Sc
11	Mot_04	Numeric	8	0	Motivation Item...	None	None	8	Right	Sc
12	Mot_05	Numeric	8	0	Motivation Item...	None	None	8	Right	Sc
13										
14										
15										
16										
17										
18										
19										

Data View Variable View

IBM SPSS Statistics Processor is ready Unicode:ON

Data Editor
window:
Variable View

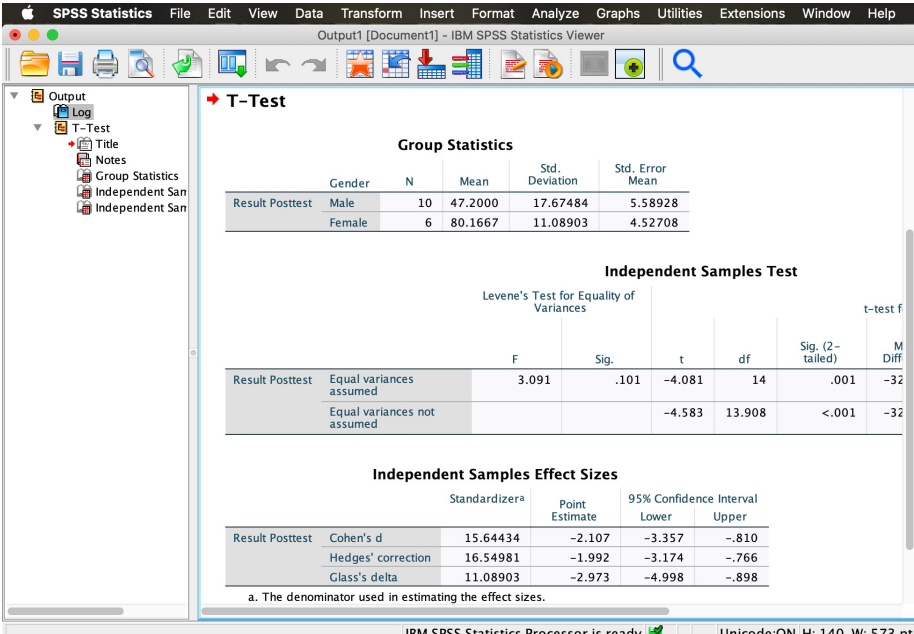
Overview of your variables and
attributes (name, scale level
(measure, labels, values, etc.)

SPSS Environment: Data Editor window

- ◇ Default window
- ◇ Displayed in spreadsheet format
- ◇ Contains 2 views: Data and Variable
- ◇ Data View: columns represent variables, rows represent cases
- ◇ Variable View: Shows information about variables present in the open data (but not the data itself)

SPSS Environment

Each SPSS window is associated with specific tasks and types of SPSS files.



The screenshot shows the SPSS Statistics Viewer window with the following data:

Group Statistics

	Gender	N	Mean	Std. Deviation	Std. Error Mean
Result Posttest	Male	10	47.2000	17.67484	5.58928
	Female	6	80.1667	11.08903	4.52708

Independent Samples Test

		F	Sig.	t	df	Sig. (2-tailed)	M Diff
Result Posttest	Equal variances assumed	3.091	.101	-4.081	14	.001	-32
	Equal variances not assumed			-4.583	13.908	<.001	-32

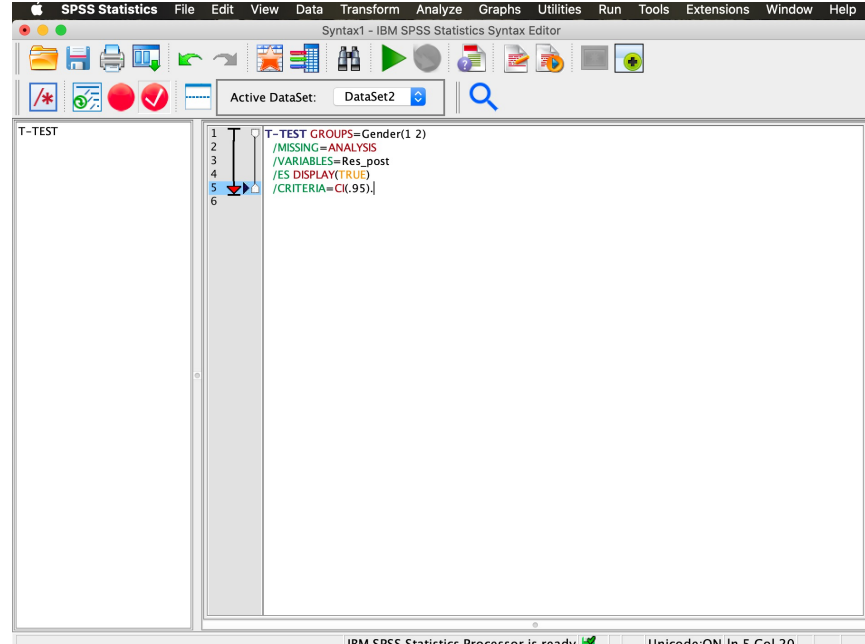
Independent Samples Effect Sizes

	Standardized	Point Estimate	95% Confidence Interval	
	Cohen's d		Lower	Upper
Result Posttest				
	Cohen's d	15.64434	-2.107	-3.357
	Hedges' correction	16.54981	-1.992	-3.174
	Glass's delta	11.08903	-2.973	-4.998

a. The denominator used in estimating the effect sizes.

Viewer
window

Executed syntax
Results from your analyses



The screenshot shows the SPSS Statistics Syntax Editor window with the following syntax command:

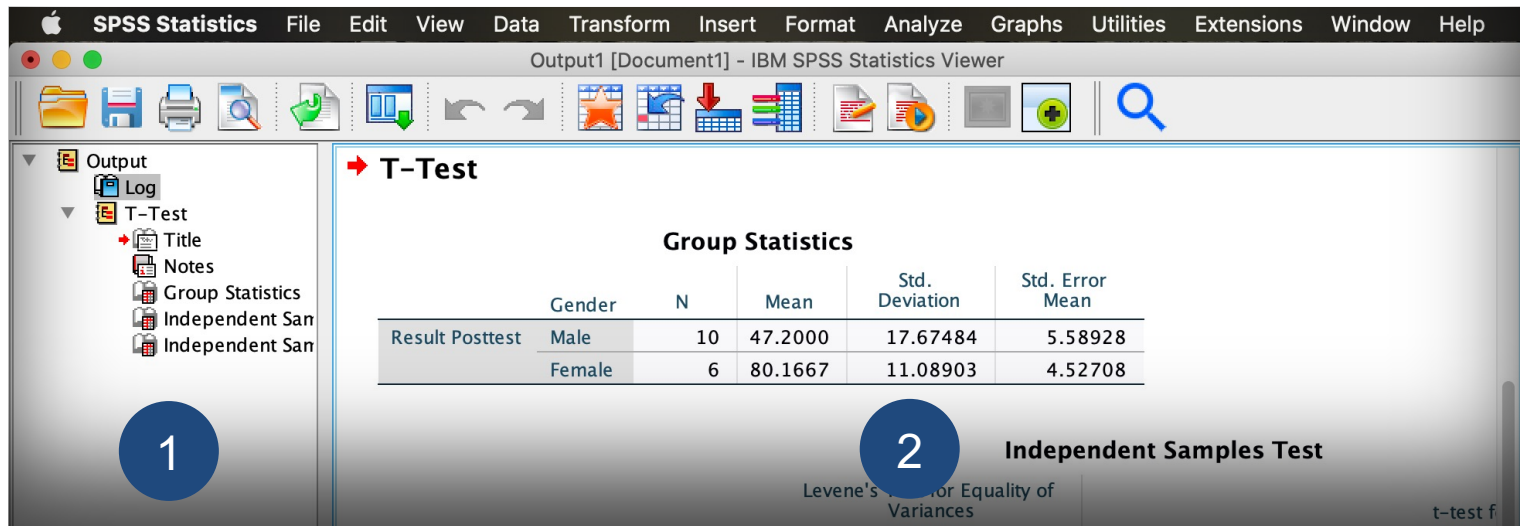
```
1 T-TEST GROUPS=Gender(1 2)  
2 /MISSING=ANALYSIS  
3 /VARIABLES=Res_post  
4 /ES DISPLAY(TRUE)  
5 /CRITERIA=CI(.95)  
6
```

Syntax Editor
window:
Variable View

Paste, edit, debug and run
syntax commands

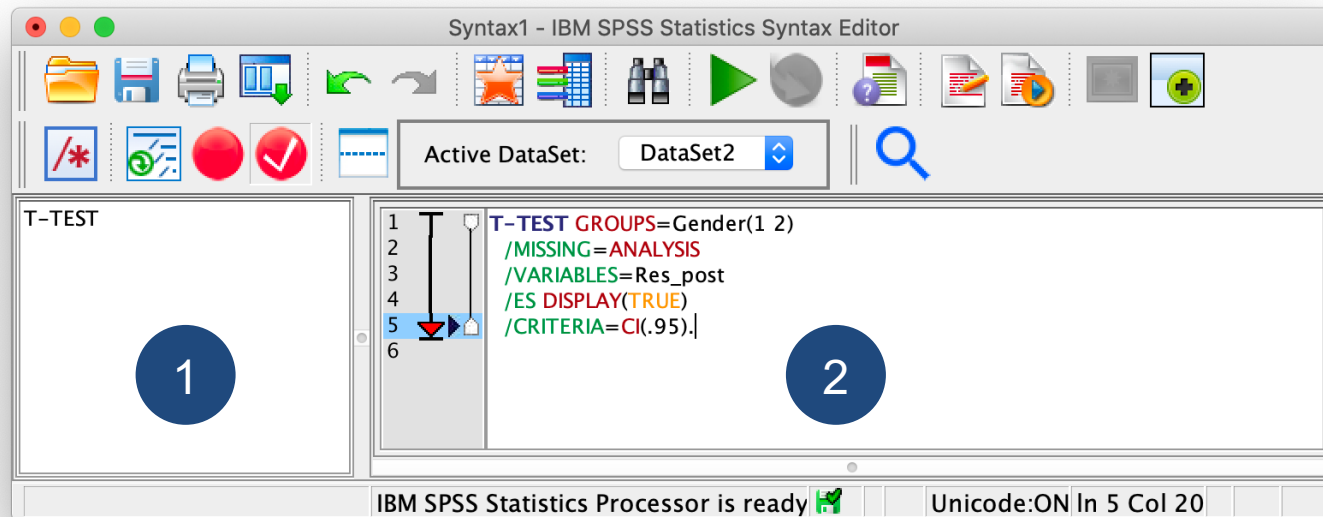
SPSS Environment: (Output) Viewer window

- ◇ Opens automatically
- ◇ Displays a log and output (syntax and results) of the actions taken as well as the result of statistical analysis
- ◇ Contains the outline of the content in the viewer (1) and the actual output (2)
- ◇ Contents can be modified and saved as a viewer file (*.spv) and other formats (e.g.: *.xlsx, *.doc, *.png)



SPSS Environment: Syntax Editor window

- ◇ SPSS syntax is a programming language (alternative to menus)
- ◇ Users can write, debug and run
- ◇ Shows an outline of commands (1) and the syntax editor (2)
- ◇ Can be saved as an *.sps file
- ◇ To open a syntax window, go to: File > New > Syntax
- ◇ To run syntax code click the “Run Selection”  button in the Syntax Editor window



SPSS Environment

SPSS affordances:

- ◇ It has a user interface → easier to learn
- ◇ Can be used for presenting results, e.g. result tables using APA formatting
- ◇ (Output) Viewer window: shows the syntax of every analysis/action. That syntax can be copied and then pasted in a Syntax Editor window
- ◇ Inclusion of add-ons (called macros) for more sophisticated analyses. Example PROCESS macro:
<https://www.processmacro.org/download.html>

SPSS Environment

SPSS disadvantages:

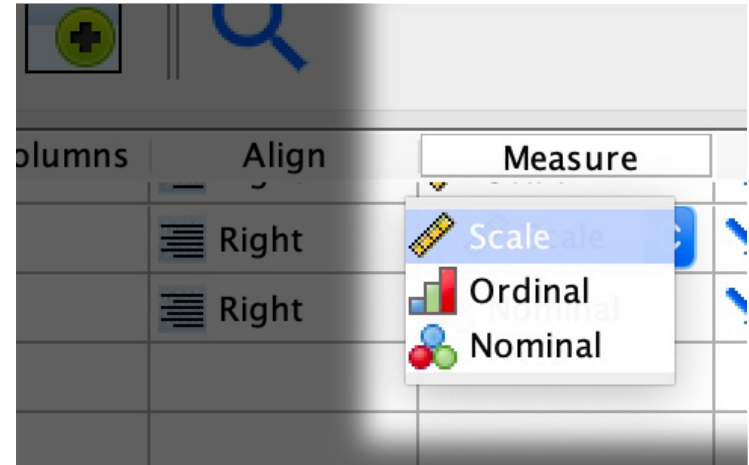
1. Not free in terms of freedom to use and modify the code (i.e. not open-source)
2. Not free in terms of money (i.e. you need pay/acquire a user license)
3. Related to 1, latest trends in statistics and plotting take a while to appear on SPSS

SPSS: SCALE LEVELS

Scale levels

Categories:

1. Nominal: for nominal data
2. Ordinal: for ordinal data
3. Scale: for interval and ratio data



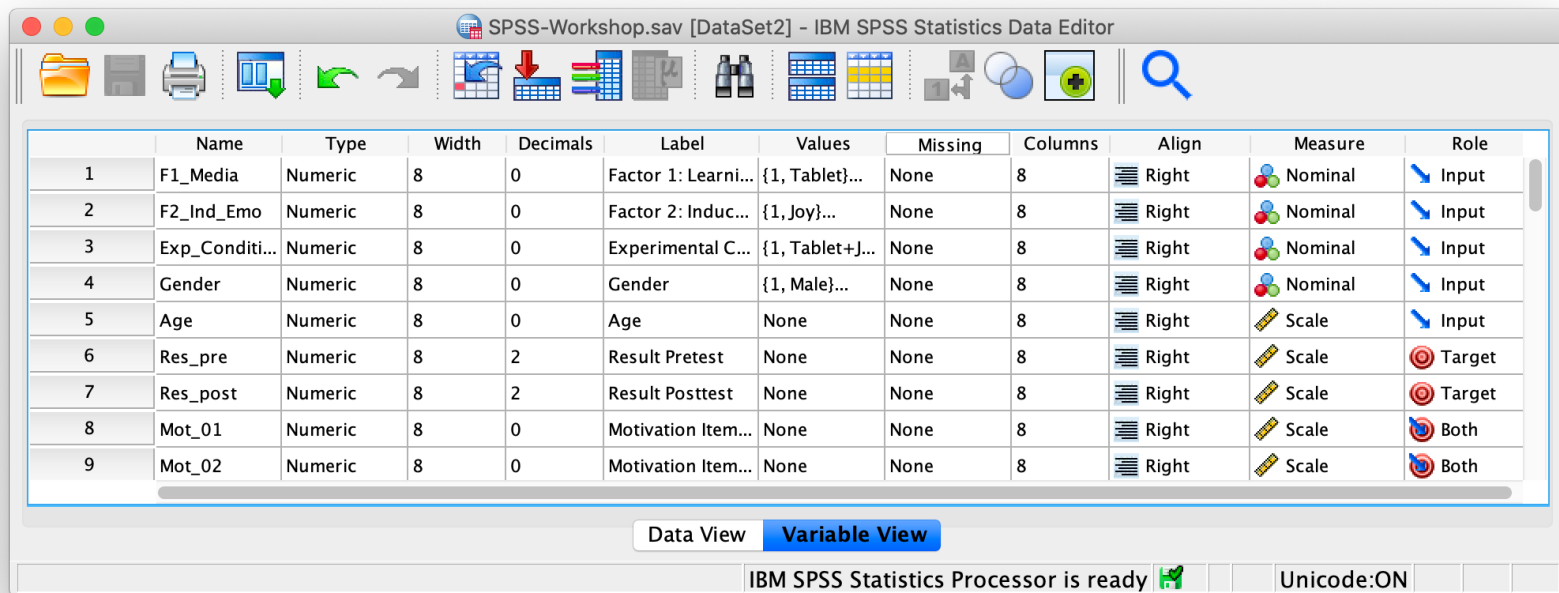
Important: Ensure to set your variables on the correct measure category!

SPSS: VARIABLES

Variables in Data and Variable view & Value labels

Variables

You can manage variables using the Variable View, once you enter the variable name, all the fields will be filled with default values that you can adjust if needed

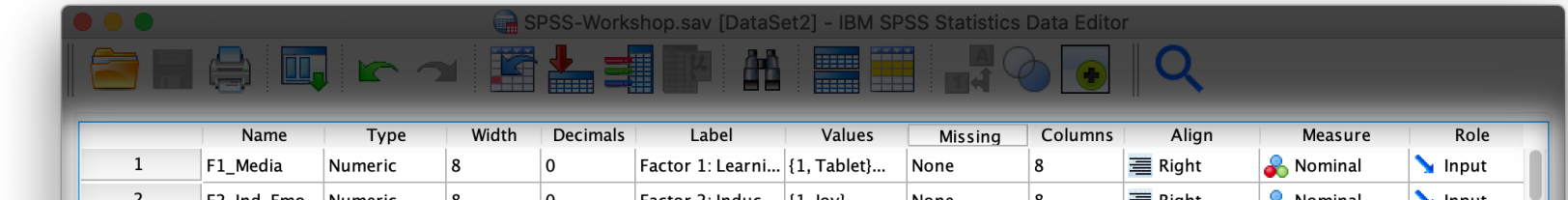


Variables

Best practices for naming variables in SPSS:

- ❖ Each variable name must be unique
- ❖ Keep variable names up to 64 characters long.
- ❖ No spaces, you may include alphanumeric characters, non-punctuation characters. Also periods and underscores within (not at the end).
- ❖ You can use upper and lower cases (Camel case - e.g. typeOfcar).
- ❖ SPSS has reserved keywords, you can't use them as a variable name (i.e.: ALL, AND, BY, NOT, OR, TO or WITH).
- ❖ You can add any type of character and quantity at the "Label", but also keep it short to keep the output more "readable".

Variables in Data View



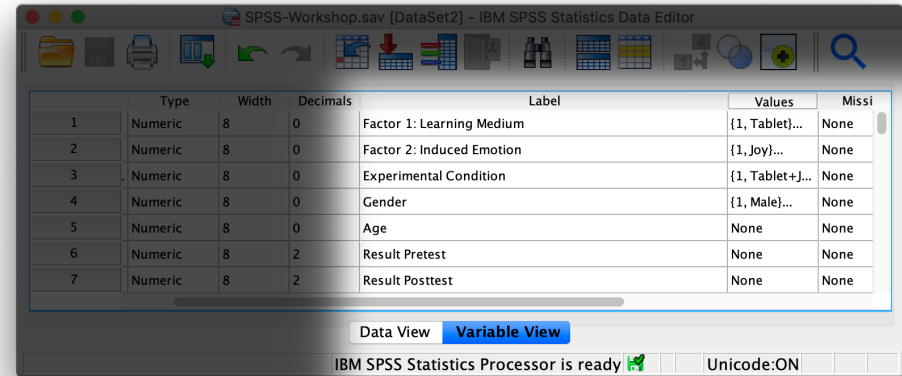
	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	F1_Media	Numeric	8	0	Factor 1: Learni...	{1, Tablet}...	None	8	Right	Nominal	Input
2	F2_Ind_Emo	Numeric	8	0	Factor 2: Induc	{1, low}	None	8	Right	Nominal	Input

- ◇ **Name:** name of the variable
- ◇ **Type:** type of variable (e.g.: string, numeric, comma, date)
- ◇ **Width:** number of digits displayed (for numeric values) or the number of characters (for a string)
- ◇ **Decimals:** number of decimal digits to be shown
- ◇ **Columns:** the width of the actual column in the Data View (not the number of digits).
- ◇ **Align:** the alignment of content of the cell.
- ◇ **Measure:** level of measurement for the variable nominal, ordinal, or scale (interval or ratio). This setting affects everything from graphs to internal algorithms for statistical analysis.
- ◇ **Role:** how the variable will be used in your analysis (e.g.: input for IV or target for DV)

Variables in Variable View

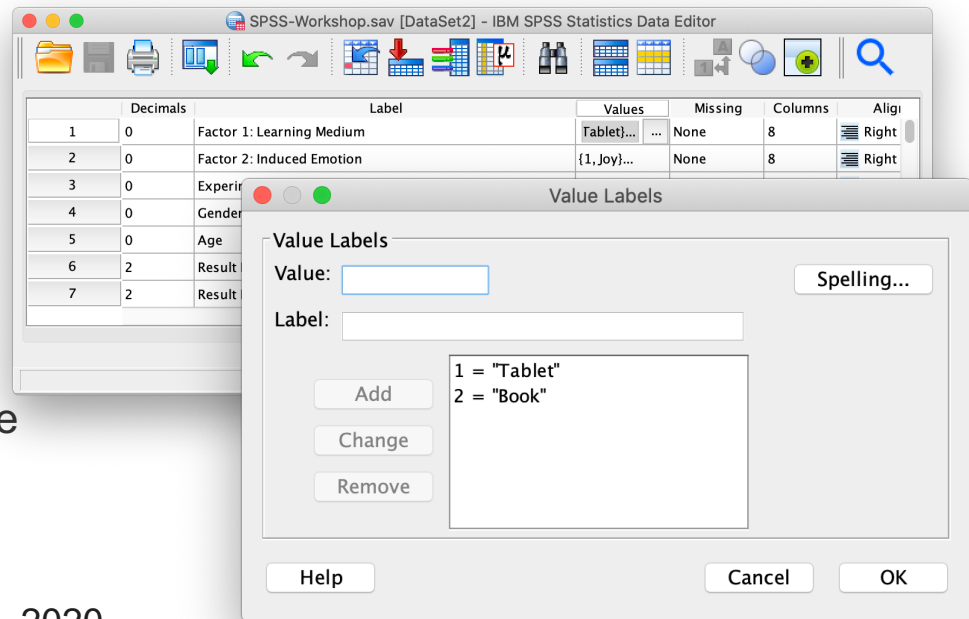
Labels:

- ◇ Brief and descriptive display name for the variable
- ◇ Labels will be shown in the output



Value labels:

- ◇ For coded categorical variables (e.g. 1 = "Tablet"; 2 = "Book")
- ◇ Helps understand what each value represents
- ◇ The value labels can be displayed in the output instead of the original numerical codes
- ◇ To edit the values of a variable click on the  button located on each cell from the "Values" column



Adapted from the slides from Claudio Alves da Silva, 2020

SPSS Environment: Variable value labels in Data View

Labels:

- ◇ After defining values, in the **Data View** you can click on the

“Value labels” button  to switch between values and their labels



	AlterCluster
6	2.00
5	1.00
8	2.00
5	1.00
1	2.00
3	3.00



	AlterCluster
6	20–29 years
5	up to 19 years
8	20–29 years
5	up to 19 years
1	20–29 years
3	30–39 years

SPSS: VARIABLES

Missing values & Define/Recode missing values

SPSS Environment: Missing values

- ◇ When can it happen? When a participant doesn't answer a question, when an answer is illegible and can't be added to the dataset
- ◇ String variables when blank are not automatically recognized as missing values
- ◇ Numeric variables when missing are attributed a default value "."

SPSS Environment: Define/Recode missing values

- ◇ We can recode a specific value to be treated as missing value data
- ◇ This new recoded missing value **must** be a value outside of the range of the real data ← Why you think that's the case?

SPSS Environment: How-to recode missing values

Recoding missing values in Data View

The screenshot illustrates the process of recoding missing values in SPSS. The main window shows a data table with columns 'vpnr', 'sex', and 'alter'. The 'Transform' menu is open, and 'Recode into Same Variables...' is selected. A dialog box 'Recode into Same Variables' is shown, with 'alter' selected in the 'Numeric Variables:' list. The 'Old and New Values...' button is highlighted. A second dialog box, 'Recode into Same Variables: Old and New Values', is shown below, with 'System-missing' selected for the 'Old Value' and 'Value: 9999' selected for the 'New Value'. The 'Old --> New:' list shows 'SYSMIS --> 9999'.

vpnr	sex	alter
1	2	20
2	3	21
3	4	27
4	5	23
5	6	21
6	7	20
7	8	20
8	9	22
9	10	26
10	11	19
11	14	21
12	15	20

Adapted from the slides from Claudio Alves da Silva, 2020

SPSS Environment: How-to define missing values

Defining missing values in Variable View

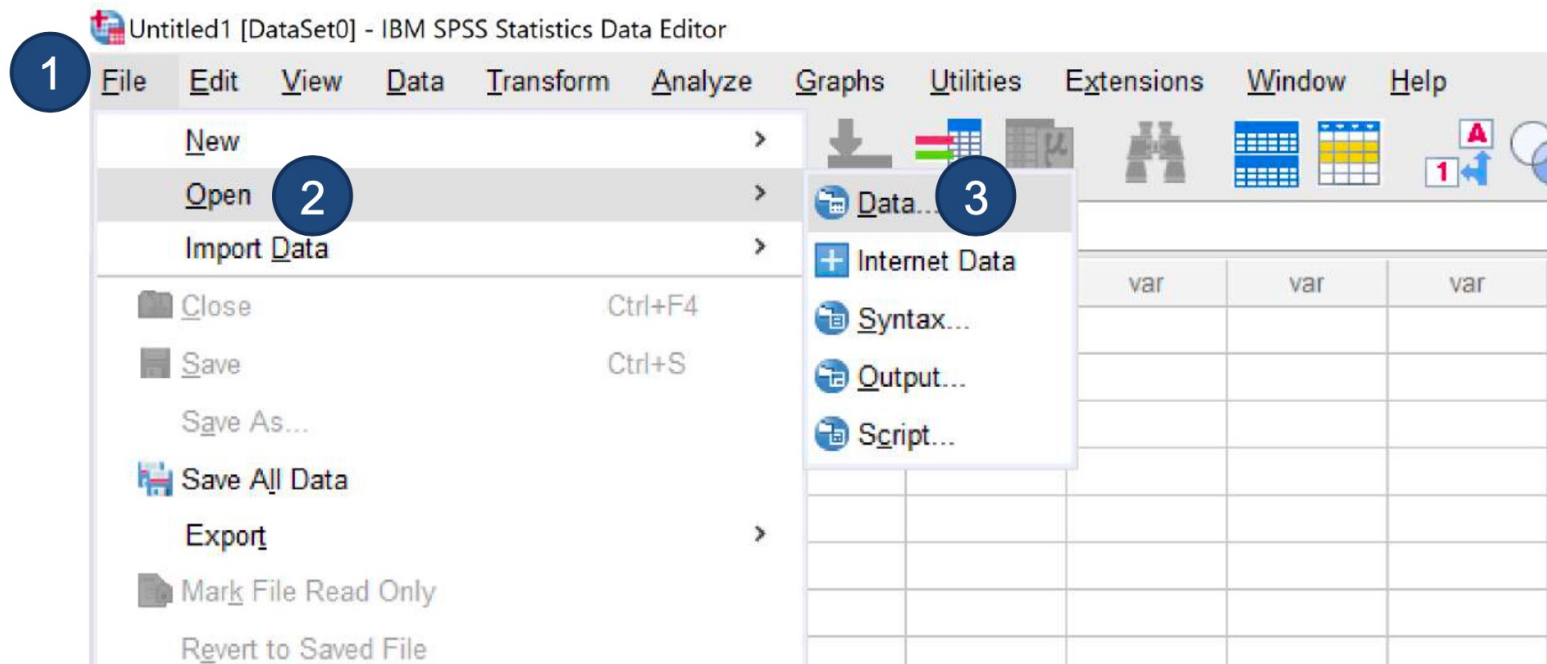
	Name	Type	Width	Decimals	Label	Values	Missing
3	alter	Numeric	2	0	Age	None	None

The diagram illustrates the process of defining missing values in SPSS. It shows a table in Variable View with columns: Name, Type, Width, Decimals, Label, Values, and Missing. The row for 'alter' (Numeric, Width 2, Decimals 0, Label Age) has 'None' in the Values and Missing columns. A blue box highlights the '...' button in the Missing column. An arrow points from this button to the 'Missing Values' dialog box. The dialog box has three options: 'No missing values' (selected), 'Discrete missing values', and 'Range plus one optional discrete missing value'. The 'Discrete missing values' option is selected in the second dialog box, and the 'OK' button is highlighted.

SPSS: IMPORTING EXCEL DATA

Importing data from Excel to SPSS

- ◆ In the Data Editor window:

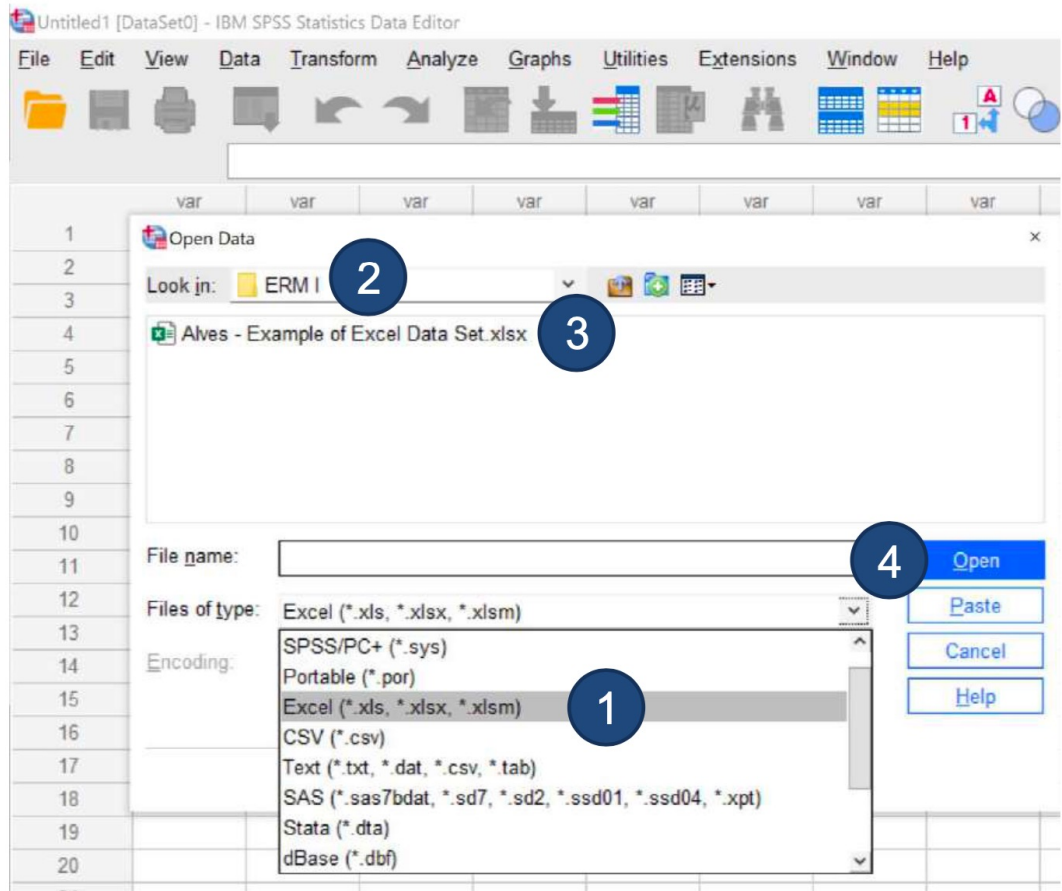


These instructions are validated for SPSS Version 25 to 27.

Importing data from Excel to SPSS

In the “Open Data” window, select the “Excel” option* (1) then navigate to the directory where the file is located (2), select the file (3) and click to open (4).

*If you do not specify the type of file that you wish to open, your file may not appear in the list of available files



Importing data from Excel to SPSS

In the “Read Excel File” window, choose the worksheet that contains your data* (1).

If you want, you can specify the range of rows / columns to import (2).

Select (3) if your variable names are in the first row of data, and (4) if you want SPSS to assign a data type to your variables**, you can also ignore hidden data (5).

The options to remove leading and trailing spaces from string values (6) removes any whitespace characters that appear at the beginning or the end of the string.

Click OK (7) and the data will appear in SPSS.

* SPSS will import one sheet at a time.

** Any number above 50.

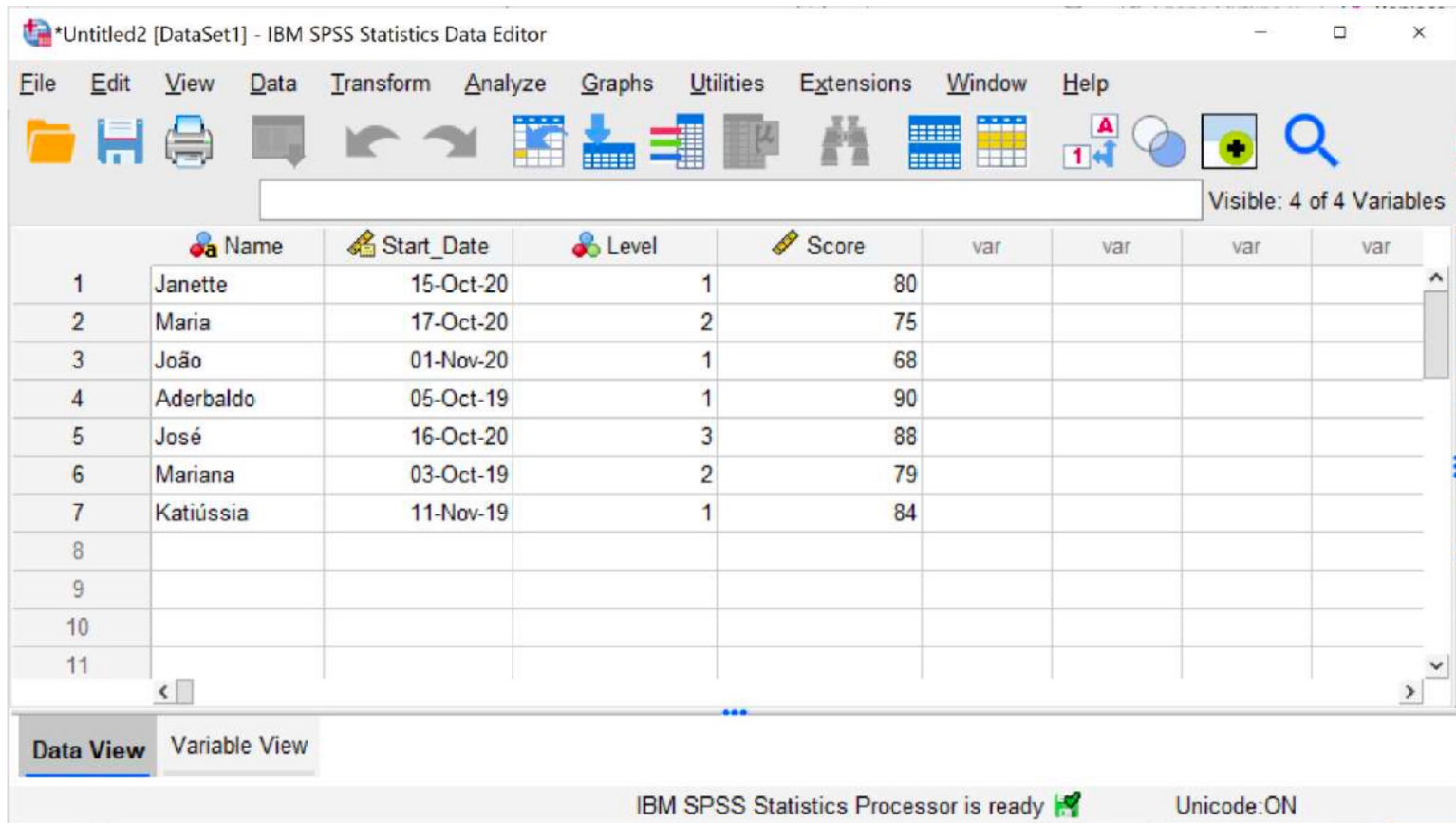
The screenshot shows the 'Read Excel File' dialog box. The file path is 'C:\Users\calve\Documents\EduTech\ERM IVAlves - Example of Excel Data Set.xlsx'. The 'Worksheet' dropdown is set to 'Sheet1 [A1:D8]' (1). The 'Range' field is empty (2). The 'Read variable names from first row of data' checkbox is checked (3). The 'Percentage of values that determine data type' is set to 95 (4). The 'Ignore hidden rows and columns' checkbox is checked (5). The 'Remove leading spaces from string values' and 'Remove trailing spaces from string values' checkboxes are unchecked (6). The 'Preview' section shows a table with 7 rows and 5 columns: Name, Start, Level, and Score. The 'OK' button is highlighted (7).

	Name	Start	Level	Score
1	Janette	10/15/2020	1	80
2	Maria	10/17/2020	2	75
3	João	11/01/2020	1	68
4	Aderbaldo	10/05/2019	1	90
5	José	10/16/2020	3	88
6	Mariana	10/03/2019	2	79
7	Katiússia	11/11/2019	1	84

Final data type is based on all data and can be different from the preview, which is based on the first 200 data rows. The preview displays only the first 500 columns.

7 OK Paste Reset Cancel Help

Importing data from Excel to SPSS



The screenshot shows the IBM SPSS Statistics Data Editor window titled '*Untitled2 [DataSet1] - IBM SPSS Statistics Data Editor'. The window has a menu bar (File, Edit, View, Data, Transform, Analyze, Graphs, Utilities, Extensions, Window, Help) and a toolbar with various icons. Below the toolbar, a status bar indicates 'Visible: 4 of 4 Variables'. The main data grid displays the following data:

	Name	Start_Date	Level	Score	var	var	var	var
1	Janette	15-Oct-20	1	80				
2	Maria	17-Oct-20	2	75				
3	João	01-Nov-20	1	68				
4	Aderbaldo	05-Oct-19	1	90				
5	José	16-Oct-20	3	88				
6	Mariana	03-Oct-19	2	79				
7	Katiússia	11-Nov-19	1	84				
8								
9								
10								
11								

At the bottom of the window, there are tabs for 'Data View' (selected) and 'Variable View'. The status bar at the very bottom indicates 'IBM SPSS Statistics Processor is ready' and 'Unicode:ON'.

Example of how data appears in the SPSS Data view window once the data have been successfully imported.

SELF-PACED ACTIVITIES

Activity 1 & 2

- Activity 1
 - Recode variable
- Activity 2
 - Compute variable